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Mixed-Reality Design for Broken-World Thinking

Kari Kraus, Derek Hansen, Elizabeth Bonsignore, June Ahn,
Jes Koepfler, Kathryn Kaczmarek Frew, Anthony Pellicone,
and Carlea Holl-Jensen

The past often only survives in fragments. The work of historians, antiquarians, and archaeologists has traditionally involved trying to reconstruct history, to put the past back together, and augmented reality is providing exciting new possibilities to assist in these efforts. Kari Kraus and her research team argue that augmented reality can also be used to explore the knowledge gained from broken, exploded, and unfinished objects. In this chapter, Kraus et al. discuss how they used these ideas in the 2015 augmented reality game DUST, and the potential these ideas hold for helping audiences see the past in new ways.

Within library, archives, and museum settings, augmented-reality (AR) projects are often motivated by an ethos of restoration and repair, resulting in an abundance of tools that digitally reconstruct cultural antiquities ravaged by time and circumstance. Such virtual reconstruction techniques have been used to recreate faces on murals, architectural details on temples, missing words from ancient manuscripts, lost shards from broken vases and vessels, and cityscapes destroyed by earthquakes, among a host of other applications.¹ The theme of *broken-world thinking* is thus central to the concerns of this chapter. Coined by Steven Jackson in “Rethinking Repair,” broken-world thinking “asks what happens when we take erosion, breakdown, and decay, rather than novelty, growth, and progress” as “start-

ing points” in our understanding of old tech and our design of new.² In this chapter we argue that the completionist tendencies prompted by AR in cultural heritage contexts—such as the digital restoration of a broken vase—should be balanced by fragmentary approaches favored by various schools of design and consistent with the historical truth of fragmentation and decay.

Based in part on Kraus’s research into how individuals identify the constituent parts of objects—including broken, obsolete, and semantically ambiguous objects—we distill a set of AR design principles addressed to historians, game designers, and library, archives, and museum professionals. We show these principles at work in two contexts: (1) A mechanical drawing of a telegraph—loosely modeled after exploded-view diagrams—that documents the material origins of Samuel Morse’s invention. The larger aim of this proof of concept is to open a window onto alternative history-enriched design paradigms for AR in science and technology museums. (2) An educational alternate-reality game focused on the deep-time sciences, which ran for two months in early 2015 (see the trailer at <http://fallingdust.com>). Funded by the National Science Foundation, DUST incorporates AR apps and 360-degree panoramas that have been conceptualized in part as *non-finito products*: deliberately unfinished things intended to be completed by users.³

Parts and Wholes: Exploded Views for Science and Technology Museums

The popularity of AR as a technique for virtual reconstruction within museums is understandable given the material barriers to restoration that otherwise exist. Unlike a damaged manuscript whose text can be conjecturally recovered in a new edition that is physically separate from the original, the mending and repair of paintings, statues, and other works of art are carried out in situ, thus potentially confounding any distinctions between the contributions of the original artist and those of the conservator. Within cultural heritage contexts, the established principle of *discernibility* has provided a practical workaround to this problem: any intervention must be visually distinct from the original and yet, paradoxically, harmoniously integrated with it.⁴ In practice this may be accomplished through a variety of means, including the application of thin, striated brush strokes known as *tratteggio*,⁵ or even by creating a recessed zone on a canvas that can function as a kind of safe harbor for experimenting with more audacious conjectures.

The promise of AR is that the art object can be *visually* altered without being *materially* modified. Researchers affiliated with the European Virtual Engineering Foundation, for example, have experimented with projecting a polychromatic digital layer over the stone sculpture of the Madonna displayed in the portico at St. Mary's Cathedral in Spain, thus aesthetically reintegrating lost color into the gothic statuary.⁶ While this approach has much to recommend it, it nonetheless risks propagating an atemporal view of cultural heritage by attempting to experientially reverse the effects of aging, privileging an originary moment over subsequent transformational states, and valuing stasis over change. In this chapter, by contrast, we are interested in exploring alternative paradigms for AR that are not bound by the logic of restoration. Inspired by Daniela Rosner et al.'s analytic category of *material traces*, we embrace ideas of provenance, breakage, wear, endurance, and fragmentation.⁷ "Time," writes Rosner, "has been under-represented in the conceptualization of artifacts and their design."⁸ Drawing on the prevalence of exploded-view 3D models and mechanical drawings in AR marketing design, particularly in the automotive sector, we adapt and reconceptualize them for a museum context.

Like a freeze frame of a detonation, an exploded diagram represents its subject as if its components were hovering in midair, torn asunder yet still perfectly preserving their order and relationships. The object is thus visually suspended between assembly and disassembly, between cohesion on the one hand and physical disintegration on the other. In this split-second time frame, individual parts preserve their allegiance to one another while simultaneously being forced apart. The exploded view is thus a generative metaphor for thinking through the dynamic role of parts in technological design, breakdown, and invention.

The initial impetus for Kraus's research study was a literature review on the subject of technological change. Historically we know that many new technologies have inherited parts from prior technologies. The skateboard remediated the surfboard; the camera pilfered from the gun; the telephone harvested batteries and wires from the telegraph; and early models of the phonograph borrowed the treadle mechanism of the sewing machine. In each of these instances, the logic of part-whole relationships governs the design. "Many of a technology's parts are shared by other technologies," notes Brian Arthur in *The Nature of Technology*, "so a great deal of development happens automatically as components improve in other uses 'outside' the host technology."⁹

To better understand this process, Kraus decided to take a closer look

at the role of parthood in creativity and design. She recruited thirty research subjects at the University of Maryland who were asked to examine six three-dimensional artifacts and complete a written questionnaire about them. These artifacts ranged from the familiar (a book) to the unfamiliar or imaginary (a 3D-printed steampunk variation on the medieval astrolabe) to the broken or visually distinctive (a fork with a bent tine and a rock that appears to have a face, a phenomenon known as *pareidolia*).¹⁰ The questionnaire asked participants to identify each object or hazard a guess about it if they didn't know what it was (*object recognition*); analyze it into its constituent parts (*component analysis*); identify the most significant, distinctive, or salient part (*significant properties*); and describe the process by which they performed these tasks and any challenges they faced (*self-reflection*).

One of the more curious findings of the study was the degree to which individuals showed tolerance for deviation from a prototype. In the case of the broken fork, for example, it was often viewed as whole and redeemable as long as—and this is key—it was also perceived as still useable. In such instances test subjects seemed to mentally normalize the object, suggesting that something like a reparative psychology was at work. Indeed, in three separate cases, the bent tine was never mentioned at all, despite its glaring conspicuousness. For those who concluded the fork was no longer functional, however, the response to the bent tine took on a very different character. In one notable instance, the reaction was spectacularly hostile and indignant: “One [tine] is BENT!! This is no longer a fork!!!! Ahhhh *hack hack.* I can't tell you how distressing it is to look at a fork that is bent out of shape like this. It is not a fork—not a fork—not a fork. . . . I had to cover it back up again. When I uncovered the ball, there was delight—the fork was distress, and it makes a horrible noise on the table. Owwww.”

“Not a fork”: for participants who reached this conclusion, the broken tine functioned as a transformational lever, allowing them to begin to see the utensil as *something else*. The distorted prong “can behave differently than a straight piece of metal,” one test subject observed. It “changes the purpose of the object,” echoed another. The fork became weaponized for one participant, presenting the possibility of “non-trivial nasal penetration”; and it morphed into a trident for someone else, that classic three-pronged spear we associate with Neptune. The invocation of the trident suggests that broken parts may be particularly susceptible to what cognitive scientist Jennifer Freyd calls “representational momentum,” in which individuals project a hypothetical future state onto an object.¹¹ In this case the test subject mentally followed the trajectory of deterioration to its logi-

cal end, specifying a path along which the tine eventually transitions from a merely bent state to a fully detached one, leaving behind something that is resolutely “not a fork.”

The tine of the fork has a counterpart in the form of the needle on the astrolabe.¹² The needle isn’t broken, but it is moveable, a feature that was often singled out as important by test subjects. The way they responded to the needle suggests that the parts of a technology most emblematic of function—of working order and design, such as the minute hand of a clock or the spout of a teakettle—are often simultaneously (and paradoxically) the parts that appear most readily breakable, snapable, and detachable.

The astrolabe, an ancient instrument used by mariners to navigate by stars and planets, provided an interesting case. Nearly 70 percent of participants—a large majority—identified the pointer as the most important part. The measuring blade came in at a distant second, with just 20 percent identifying it as most important. And tied for last place were the stem and eyepiece combo and the cross-hairs, each of which received just under 7 percent of the vote. Test subjects tended to use the pointer diagnostically in an attempt to figure out what the astrolabe was. These diagnoses were conjectural and nearly always inaccurate, but governed by an analogical imagination that lends them plausibility. Like the broken tine of the fork, the pointer triggered the invocation of other objects, including a game spinner, a compass, a clock, an old-timey elevator-floor indicator, a sextant, a gauge for measuring rain, and a steamboat propeller. Seen as the most salient part of the astrolabe, it was also conceptually detached from it, making it cognitively available for grafting onto and visualizing in other objects.

When Kraus presented on this research at the AR workshop in fall 2014, Bill Turkel insightfully remarked that knowing when to suppress the urge to repair may be vital to any culture of invention. Building on Turkel’s point, we’d add that the ability to locate the fault lines in an object, to reason combinatorially with its parts, to conceptually move those parts into and out of different artifactual contexts, allows us to creatively make and unmake the world.¹³

Consider the telegraph as a case study. In 1844, Samuel Morse, a one-time professor of arts and design at New York University, sent a sequence of electromagnetic pulses over wire from Washington, D.C., to Baltimore, Maryland. Those pulses transmitted the first message sent via telegraph. The content of that first coded message—“What hath God wrought!”—was chosen by Annie Ellsworth, the daughter of Morse’s friend.¹⁴ If there was a current of divinity barreling through that wire, as

Ellsworth's choice implies, then there was also one of common humanity: Morse's original telegraph was cobbled together from odds and ends lying about his artist's studio and his brother's print shop. Type slugs, an artist's canvas stretcher, wire from a paper mould, a compositor's stick, and old clock-works were salvaged from the jetsam to create a device that would chart a new communications course for civilization.¹⁵ The telegraph was assemblage art, hacked together from found objects. Or call it sampling and remixing. However one chooses to label it, it is the very banality of Morse's act that makes it what it is: a parable of invention.

Drawing inspiration from this assemblage model of design, we have mocked up an experimental mechanical drawing of a telegraph (figure 4.1), with the aim of suggesting how such models in general, rendered in 2D or 3D when a user captures a museum artifact on her video display, might be implemented as a mobile AR app in a science or history museum. In addition to showing the relationship of the different parts of the technology, as is characteristic of exploded-view diagrams, we have also sketched in shadowy representations of the antecedent technologies from which they inherit components. These ancestral ghosts in the machine give the lie to ex nihilo theories of invention and creativity. They also serve as a form of what has been called "material citation," visually documenting the provenance of the parts of the telegraph that make up the whole.¹⁶ Within the field of evolutionary biology, phylogenetic trees are used to visualize such patterns of descent with modification; one potential advantage of our design is that it represents not only *sources*, but also *units* of inheritance (e.g., the pendulum of the clock, but not the clock face, suspension spring, weight arm, or other mechanism). It thus serves as an affirmation of Tim Ingold's dictum that "to understand materials [and artifacts] is to be able to tell their histories."¹⁷

Non-Finito Design in an Educational AR Game

A joint endeavor between Brigham Young University and the University of Maryland in partnership with NASA and Tinder Transmedia, DUST (figure 4.2) is an alternate-reality game designed to appeal to youth aged thirteen to fifteen. Sponsored by the National Science Foundation, the game gives teens the opportunity to learn and apply STEM principles in an informal learning context. DUST is focused on the deep-time sciences: those sciences—including astronomy and evolutionary biology—that deal with processes that occur over millions or billions of years, such as the

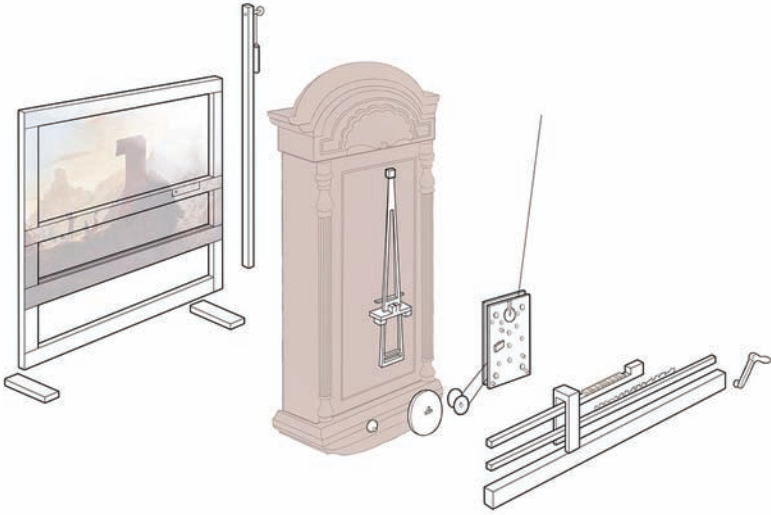


Figure 4.1. Mechanical drawing showing the component parts of Morse's telegraph. A painting, grandfather clock, and letterpress type have all been ghosted in using a brown-colored wash to suggest the source artifacts and domains from which Morse harvested parts. The oil painting—one of Morse's own—is titled *The Chapel of the Virgin at Subiaco* (1830). Illustration of Morse's telegraph by Lisa Kong; Copyright Lisa Kong. Reproduced by permission. This illustration does not fall under a Creative Commons license.



Figure 4.2. DUST poster image. (Copyright 2015 BYU. Reproduced by permission. Artist credit: Melissa Manwill. Released under a CC BY-NC-ND license. <https://creativecommons.org/licenses/by-nc-nd/4.0/>.)

formation of the galaxies, the evolution of species, or the continental drift of the earth. DUST is thus a “Big History” game whose fictional events unfold not in human time (measured in years, decades, or centuries), but in cosmic or geologic time (measured in eras, eons, and supereons).

DUST ran live for nine weeks, from January to March 2015, and a replayable version is currently undergoing play-testing. The game centers on the mysterious collapse of adults worldwide who fall into a coma-like state following a cataclysmic meteor shower. Players “hack” into NASA research databases and engage in collaborative play and inquiry across multiple media platforms to search for answers that will save their parents’ lives. Eventually they discover that the meteor dust is full of microscopic extremophiles whose DNA contains the records of a lost alien civilization, a voice from the DUST, whose world was destroyed. It is these alien microbes that have inadvertently (or perhaps purposely) “infected” the adults, knocking them out, and leaving traces of these records in their brains. By painstakingly reconstructing the ZNA of an alien species, scrambled in the act of genetic transmission, players simultaneously repair the fragmented narrative, scattered—like all good transmedia fictions—across multiple media.

While the game is overtly fictional—encouraging players to inhabit the possibility space of the “what-if”—it nonetheless offers the opportunity to engage with authentic scientific principles and theories, including the chemistry of life, the properties of habitable worlds, deep time and planetary time scales, and scientific instrumentation, such as receivers and antennas, for detecting and analyzing cosmic signals. Thus the fictional narrative serves as a compelling hook to attract teens who otherwise may have no interest in engaging with such scientific phenomena.

DUST’s AR apps are intended to help embed learning activities into the story world and provide tools with which teens can explore the fundamentals of scientific inquiry, such as data collection and analysis. Several of the apps trade on the notion of DNA as a long-term storage medium for archival records. The microscanner, for example, allows players to scan their real-world environments for (fictional) microorganisms, including the alien microbes whose ZNA contains messages to posterity. Two complementary tools—the microbe app and the Dream Decoder—give them the tools to (1) genetically engineer tiny frankencreatures capable of fighting off the aliens by manipulating nucleotide and protein sequences, and (2) decrypt rogue ZNA fragments incorporated into the adults’ native genome by the microbes, revealing the enigmatic message. The health scanner app, the second of DUST’S two mobile AR apps, simulates fMRI

technology, which can be used to perform brain scans, revealing sites of alien microbe infestation, later cured by the supertardigrades genetically engineered by players.

DUST also includes 360-degree panoramic environments created with the Unity Web Player plug-in. There are three simulated 3D environments for players to explore: the International Space Station, a NASA science lab, and an astrobiologist's office. Although these environments began as austere, seemingly uninhabited spaces outfitted with only the most basic supplies, office furniture, and lab equipment, DUST undergraduate codesigners at Brigham Young University and the University of Maryland created "artifacts" for them that helped flesh out the game's characters, themes, plot points, and scientific principles. Acting as a cross between interior decorators and mad scientists, they produced posters for walls and ceilings; books for bookshelves; screen savers, emails, and other documents for the computer monitors; voice mail and audio dictation for cellphones; and lab specimens for workstations.

While each of the environments is born-digital, many of the artifacts populating them began life as physical objects that were digitized. *Spike Field*, for example, is a creative reinterpretation of a concept design originally published in a report commissioned by the Department of Energy in 1993 on warning future civilizations about the hazards of radioactive waste.¹⁸ Created by a talented undergraduate,¹⁹ the new work adorns the wall of the office environment, which has been loosely inspired by the real office of an astrobiologist who served as a consultant on DUST and was a member of the scientific team responsible for the original *Spike Field* design. The artist used pencil to sketch and draw the layout of the scenery, followed by pastel pencils for blending and charcoal and sharpie markers for outlining. The physical work was then digitized for inclusion in the virtual environment.

Like the AR mobile apps, the 3D panoramic environments represent a combined view of physical and virtual reality.²⁰ But whereas the mobile apps use the analog world as a canvas on which to overlay digital information, the panoramic environments use a digital world on which to overlay analog information, such as the artist's drawing. The mobile and web apps thus function as inverse types of mixed reality.

In broad strokes, DUST can be understood as a *non-finito product*. Described in the research and design literature as an unfinished work meant to be completed by users, non-finito products are valued for their ability to open up a design space for user creativity.²¹ Unfinished, fragmented,

sketched, and ruined works have all traditionally fallen within the purview of the non-finito.²² Examples include a novel without an ending (unfinished work); the bust of a statue missing arms or head (fragmented work); a loosely executed drawing not intended as the basis for a final work (sketch); and a castle fallen into oblivion and decay (ruin). Recently the field of human-computer interaction (HCI) has borrowed and adapted the term, extending its reach and influence into the domains of software development and interface design. Within this larger context, Facebook’s “poke” feature has been cited as a quintessential example of non-finito design.²³ Ambiguous and open-ended, it depends on users to establish its purpose and conventions.²⁴

ARGs almost by definition fall squarely within the category of the non-finito. Players must work collaboratively to piece together and advance an adaptive narrative that is integrated into real-world media and spaces, including museums, social media sites, print novels, text messages, and the web. The features that make DUST a non-finito product can be found at all levels of design. Each week a new graphic novel chapter—or “story beat”—is released and accompanied by a call to action that players must respond to before the release of the next story beat, such as posting hypotheses, adding evidence, and collecting and analyzing data. Players have also initiated their own projects in the form of spreadsheets, Google forms, and—in one instance—a computer program for calculating star temperature and mass, all in the service of the game. In addition to being non-finito, DUST is an open-ended ARG, meaning that not every detail about the plot or storyline has been scripted in advance. Some of the best-known education, public media, and museum-focused ARGs have been open-ended, including *World Without Oil*, *Urgent Evoke*, *PHEON*, and *Superstruct*. The designers deliberately cede some control and authority to players, whose contributions help determine the shape and character of the game, as well as final outcomes. DUST’s AR apps and panoramic environments are at the heart of realizing this open-ended vision and fostering user creativity.

Central to DUST’s approach, for instance, are clearly delineated units of information that players can contribute or systematically manipulate. In the case of the microbe experimentation app, players combine, rearrange, and mutate the chemical letters of DNA to produce new forms of microbial life capable of attacking the invasive species affecting the adults. Taken together, the microscanner and experimentation apps reinforce through design the lesson that the genotype (the “inside data” as IRIS, DUST’s resident NASA AI, explains it) codes for the phenotype (the “outside phe-

nomena”). Every living organism, in other words, is “the outward physical manifestation of internally coded, inheritable information.”²⁵ Through combinatorial creativity, players thus help bring resolution to the game by creating the circumstances under which the adults can be woken up.

The mobile and web apps also work in conjunction with each other to structure player experience across media channels. Data collected through the microscanner, for example, is analyzed and manipulated using the microbe app. The player is thus pulled across the so-called “Reality-Virtuality Continuum” as she engages in scientific inquiry:²⁶ a view of the real world is first captured by the player’s smartphone camera and then overlaid with fictional information about the microbes putatively lurking on objects, humans, and animals in the environment. These microscopic organisms are then transferred to the web app, where players analyze and genetically modify them. The progression from reality to mixed reality to virtual reality thus roughly correlates with the different stages of scientific investigation (see note 17 for a definition of *virtual reality* as used in this chapter). Each stage, incomplete in itself, is one step closer to completion when combined with the next stage. While the process is decidedly iterative rather than linear, there is an overall forward momentum. As data is passed from one domain of reality to the next, its value is enhanced, its meaning enriched, its secrets gradually revealed. Players eventually decrypt it with the help of the Dream Decoder, a web app introduced in the final stage of the game. The Dream Decoder lies at the far end of the reality-virtuality continuum, the culmination of all the data collection and experiments performed further upstream with the help of the other mixed-reality apps. Taken together, the mobile and web apps are what allow players to collectively unravel the mystery.

In the case of DUST’s intentionally unfinished 3D environments, the contributions take the form of curated artifacts that add texture and realism to the story world and expand it in whimsical, suspenseful, and potentially unforeseen directions. We have also included user-created content in the environments as a way to recognize player achievements and contributions. Two long-term thinking activities introduced by Violet Cannon, one of our fictional protagonists, for example, resulted in drawings and sketches of warning systems to posterity, one of which was featured on the astrobiologist virtual office wall.

The artifacts might also function as clues or red herrings, depending on what is produced and how the artifacts are received and interpreted. In 1979, Kit Williams, author of the best-selling treasure-hunt book *Mas-*

querade introduced the term “confirmers” to denote a technique whereby multiple clues in an ARG all point to the same solution, thereby assuring players they are “on the right track and following more than a string of chance coincidences.”²⁷ The strategically placed poster and separate infographic on cryptobiosis in the ISS environment function in just this way, reinforcing the message that the hibernation-like state some extremophiles enter into under adverse circumstances is critical to solving the mystery of DUST. By obliquely cross-referencing other mentions of cryptobiosis in the graphic novel and co-lab posts, these artifacts collectively assert the importance of the scientific concept to the game.

Coda

DUST is, both literally and figuratively, a fractured world that requires players to intervene in the molecular machinery of life to salvage it. By piecing together genetic codes, players by extension stitch the seams of the story world. This *seamful design*, to borrow a term from Matthew Chalmers and Ian MacColl,²⁸ creates a unified mixed-reality composition whose sutures are intentionally visible. Like the patchwork design of the telegraph, they accentuate the relationship of parts to parts, drawing attention to sites of player creativity, engagement, invention, and inquiry.

In Japanese culture, *kintsugi* is an ancient method of repairing broken pottery using a special adhesive mixed with gold. This “golden joinery” emphasizes flaws and cracks rather than attempting to disguise them. The idea behind *kintsugi* is that the imperfections in everyday objects bear witness to human and environmental activity. By dusting them with gold, artists preserve the material traces of the past and affirm their cultural value. In this chapter we have similarly advocated for a design paradigm that enhances fault lines, proposing digital overlays inspired by exploded-view diagrams and alternate-reality games that draw on an aesthetics of fragmentation. Rather than trying to repair reality with virtual reality or invisibly stitch them together, mixed-reality design for broken-world thinking invites designers, viewers, and players to fill the cracks and seams with gold.

Notes

The short descriptions of Kraus’s mereology study and the plot of DUST draw on language Kraus and the research team have used in other talks and publications. We thank the students who participated in the mereology study and the students,

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3. See Jin-min Seok, Jong-bum Woo, and Youn-kyung Lim, "Non-finito Products: A New Design Space of User Creativity for Personal User Experience," *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (New York: ACM, 2014).

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5. For a description, see Magdalena Grenda, "Tratteggio Retouch and Its Derivatives as an Image Reintegration Solution in the Process of Restoration," *CeROArt* 2010, <http://ceroart.revues.org/1700>

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9. W. Brian Arthur, *The Nature of Technology: What It is and How It Evolves*, (London: Penguin Group, 2009).

10. We thank students in Kraus's 2013 Classification Theory course, who participated in an informal version of this study as a class activity and made recommendations on what types of objects to present to test subjects, including the suggestion that Kraus include an object exhibiting the characteristics of pareidolia.

11. Jennifer Freyd, "Dynamic Mental Representations," *Psychological Review* 94 (1987): 427–38.

12. To view the 3D-printed astrolabe by the artist Whystler that was used in the study, visit <https://www.shapeways.com/product/FBS8F8973/simple-astrolabe>

13. For a discussion of how counterfactual thinking entails the discovery of "fault lines in reality," see Ruth Byrne, *The Rational Imagination: How People Create Alternatives to Reality* (Cambridge, MA: MIT Press, 2007), 5.

14. Alexis C. Madrigal. "The First Long-Distance Telegraph Message, Sent This Day in 1844: 'What Hath God Wrought?'" *The Atlantic*, May 24, 2013. Web, accessed Oct. 7, 2015.

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19. See <https://goo.gl/48xXyz> for a reproduction and artist's credit.
20. In this chapter, we are using the term "virtual reality" to refer to online simulated environments—such as DUST's 3D panoramic models—rather than simulations that rely on special electronic equipment such as VR headsets or gloves.
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